

Module 3 Milestone Two: Narrative

Fire Department Event Tracker Application

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Artifact Description

The artifact selected for enhancement is an Android mobile application called the Fire Department Event Tracker. This application was originally created during earlier coursework in the Computer Science program and was designed to allow users to create, manage, and track fire department events. The application includes user login functionality, event creation and editing, SQLite database integration, RecyclerView event displays, and SMS notification support.

The purpose of the application is to provide a centralized event management system for fire department scheduling and communication. Users can create department events such as drills, meetings, training sessions, and parades while also viewing and managing upcoming scheduled activities.

I selected this artifact because it demonstrates several important software engineering concepts including user authentication, database management, user interface design, data validation, mobile application development, and event-driven programming. This project also closely aligns with my real-world experience in the fire service as Chief of Department for a Volunteer Fire Company.

Enhancements Implemented

Multiple enhancements were implemented to improve the overall usability, functionality, maintainability, and professional appearance of the application.

User Interface and User Experience Enhancements

The application interface was redesigned using Material 3 design principles to modernize the overall appearance of the application. Material 3 is Google's modern Android design framework. It is used as a standard to create visually consistent, accessible and responsive mobile application interfaces/experiences. A consistent red fire department themed color palette replaced the original default purple Material theme throughout the application.

Additional user experience improvements included:

- Improved spacing and layout consistency
- Rounded Material styled buttons
- Improved typography and hierarchy
- Enhanced RecyclerView event cards
- Better visual organization of event details
- Improved event detail formatting using multi-line displays
- Empty state handling when no events are scheduled
- Material 3 dark mode support
- Cleaner event scheduling layout

A custom fire department themed logo was also created and integrated into the application branding.

Event Scheduling Enhancements

The original application only supported event names, dates, and locations. The application was enhanced to include event time scheduling support.

Enhancements included:

- Added event time input field
- Added event time validation
- Updated SQLite database schema to support time storage
- Updated RecyclerView event displays to include time information
- Added formatting methods to standardize time presentation
- Improved event display readability

The final application supports properly formatted event scheduling including dates, times, and locations.

Input Validation and Error Handling Improvements

Many improvements were made to input validation and overall application stability.

Validation enhancements included:

- Username validation
- Password validation
- Empty field validation
- Date format validation using regular expressions
- Time format validation using regular expressions
- Prevention of invalid event dates
- Prevention of past event scheduling
- Improved user error messaging
- Better focus handling for invalid fields

Additional improvements were implemented to correctly allow events scheduled for the current day while still preventing events from being created in the past.

Database and Architecture Improvements

The SQLite database architecture was enhanced to support the new event scheduling functionality.

Enhancements included:

- Database schema modifications
- Added time column support
- Updated insert queries
- Updated update queries
- Updated event retrieval queries
- Improved RecyclerView adapter formatting
- Cleaner helper methods for validation and formatting
- Improved code organization and readability

The application architecture was also improved by separating functionality into reusable helper methods, improving maintainability and readability.

Course Outcomes Demonstrated

This artifact demonstrates multiple course outcomes from the Computer Science program.

Software Design and Engineering

The project demonstrates software design and engineering principles through:

- Modular application architecture
- User-centered interface design
- Database integration
- Input validation
- Error handling
- Mobile application development
- Maintainable code organization
- Event-driven programming

The enhancements improved both the functionality and maintainability of the application while also improving the overall user experience.

Algorithms and Data Structures

The application demonstrates algorithms and data structure concepts through:

- RecyclerView implementation
- SQLite database queries
- Event list management
- Input validation logic
- Pattern matching using regular expressions
- Date comparison logic

Databases

The artifact demonstrates database concepts through:

- SQLite database integration
- CRUD operations
- Schema modifications
- Data persistence
- Query handling
- Database helper implementation

Security and Reliability

The project demonstrates security and reliability concepts through:

- Login validation
- Input validation
- Runtime permission handling for SMS functionality
- Error prevention
- Date and time validation
- Defensive programming techniques
- User input sanitization
- Improved error messaging

Challenges Encountered

Several challenges were encountered during the enhancement process. One major challenge involved implementing date validation while still allowing events scheduled for the current day. Initially, the application incorrectly treated events created for the current date as past events. This issue required additional date comparison logic using Calendar objects and resetting the time values to midnight to properly compare dates without considering the current time of day.

Another challenge involved maintaining consistent Material 3 styling across both light mode and dark mode themes. Several interface elements required additional theme-aware color configuration to ensure proper contrast and readability.

Additional challenges included:

- Updating the SQLite database schema to support event times
- Maintaining RecyclerView compatibility after database changes
- Properly formatting event times
- Ensuring layout consistency across all input fields
- Handling validation edge cases

These challenges provided valuable experience in debugging, application testing, user interface refinement, and Android application architecture.

Use of Artificial Intelligence Tools

Artificial intelligence tools were used during portions of the enhancement process as development assistance tools.

AI-assisted support was used for:

- Generating ideas for the custom fire department logo design
- Assisting with troubleshooting date validation logic
- Debugging issues related to allowing current-day events while preventing past dates
- Improving Material 3 theme configuration
- Assisting with UI refinement ideas and formatting improvements

The use of AI tools functioned similarly to a software development assistant by helping identify logic issues, propose solutions, and accelerate debugging and refinement tasks. All implementation decisions, testing, validation, and final integration were completed by me.

What I Learned

This enhancement process strengthened my understanding of Android application development, software engineering principles, and user interface design.

Key lessons learned included:

- Material 3 Android design principles
- Android theme management
- RecyclerView customization
- SQLite schema modification
- Input validation techniques
- Date and time validation handling
- Defensive programming techniques
- Mobile application usability improvements
- Software debugging and testing strategies

The project also improved my ability to refactor existing code while maintaining application stability and functionality.

Conclusion

The Fire Department Event Tracker application was successfully enhanced to improve functionality, usability, maintainability, and overall user experience. The implemented

enhancements significantly modernized the application while also improving reliability and user interaction.

The final artifact demonstrates multiple software engineering competencies including mobile development, database integration, user interface design, validation logic, debugging, and maintainable architecture. The project also demonstrates my ability to improve an existing software artifact using iterative enhancement techniques and modern Android development practices.

The completed enhancements resulted in a more professional, functional, and user-friendly mobile application suitable for demonstrating software design and engineering skills within the Computer Science capstone course.